

Lecture 1: Introduction to “Computer Vision”

Harbin Institute of Technology (Shenzhen)

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Today's agenda

- What is computer vision
- Course overview



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A picture is worth a thousand words



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A word is worth a thousand pictures



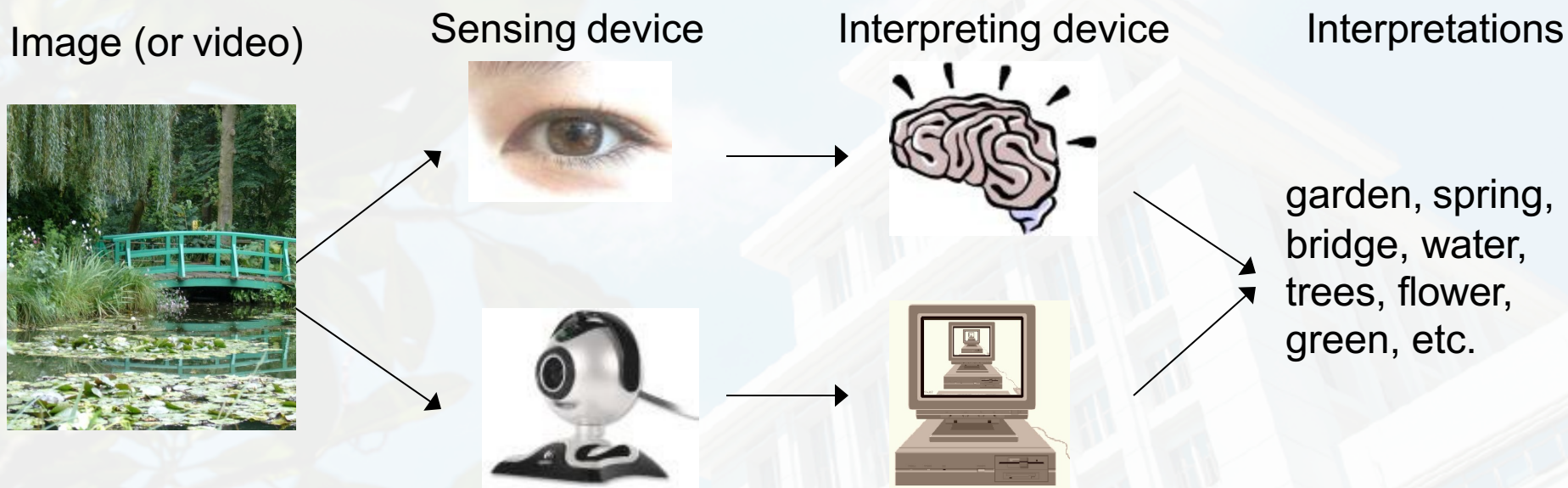
A HUNT



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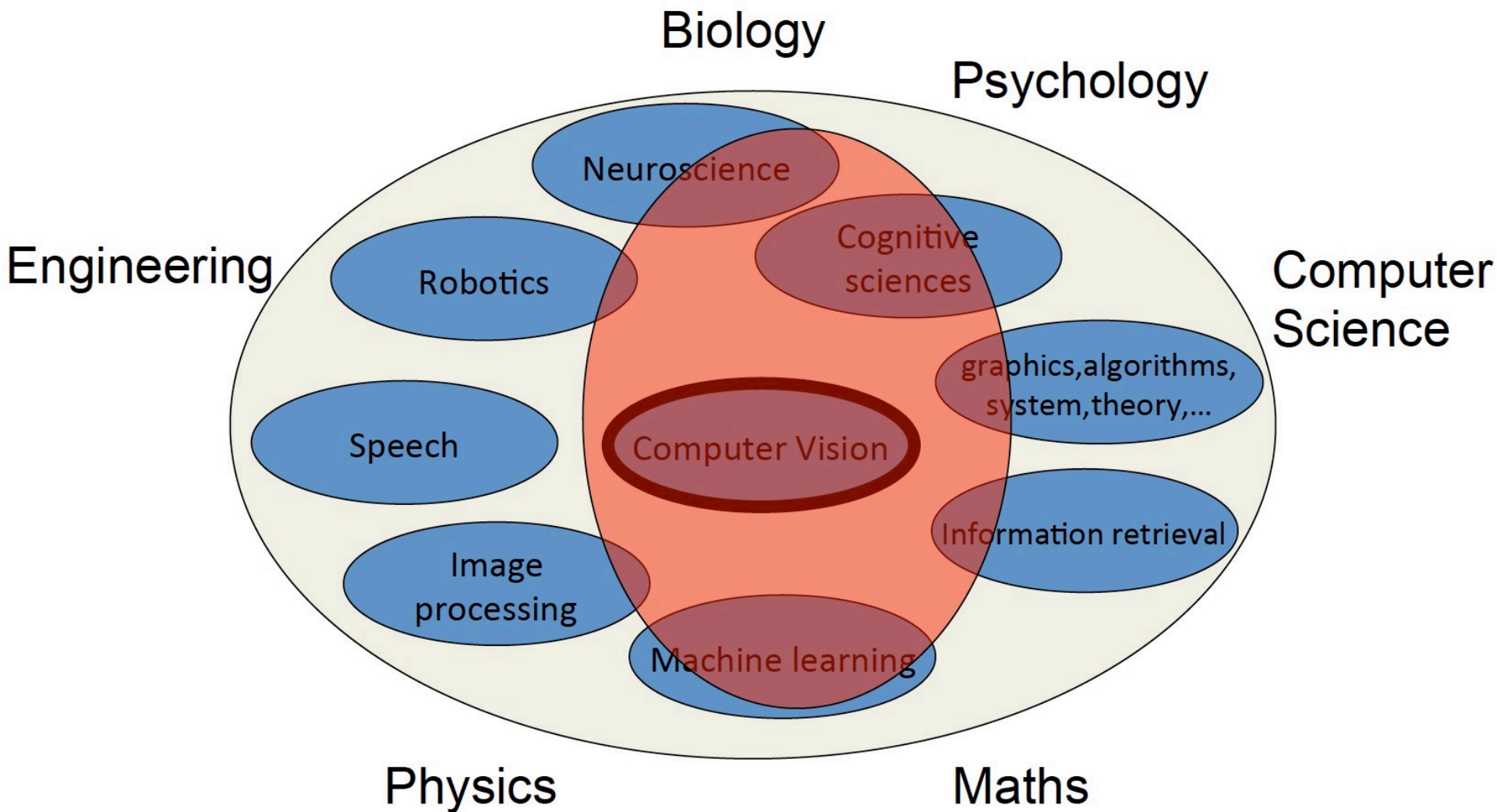
What is (computer) vision?



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What is related to?



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The goal of computer vision

- To bridge the gap between pixels and “meaning”



What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

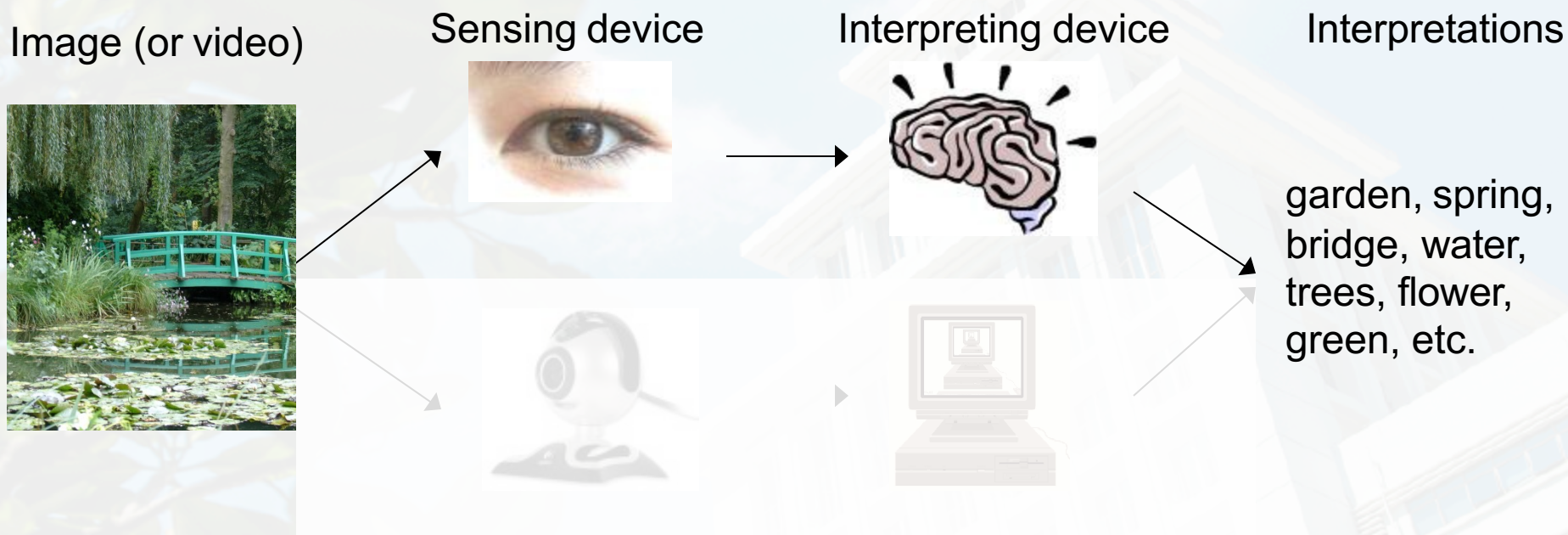
What a computer sees



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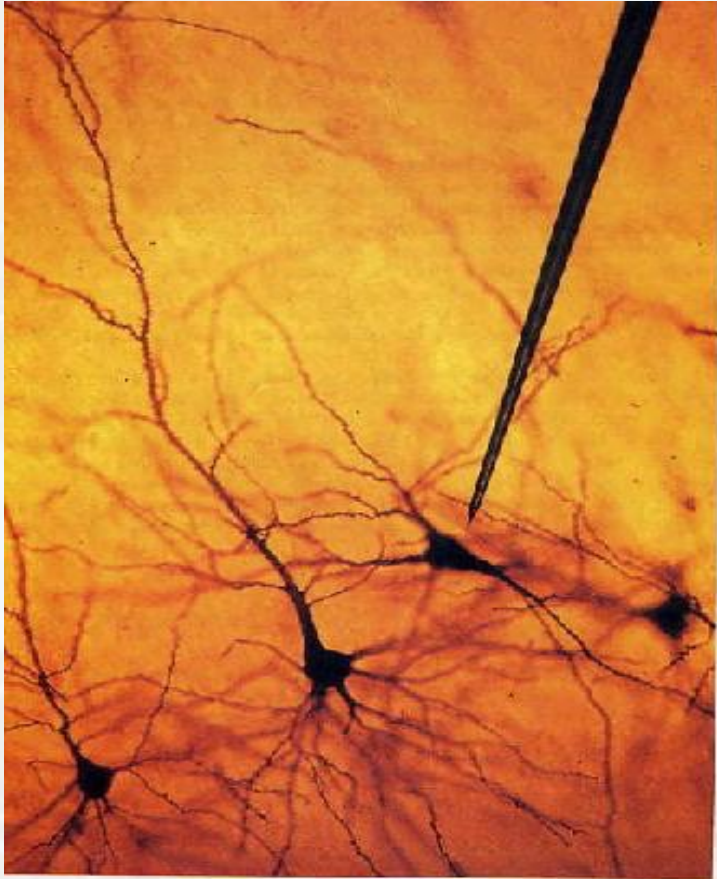
What is (computer) vision?



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1981: Nobel Prize in medicine

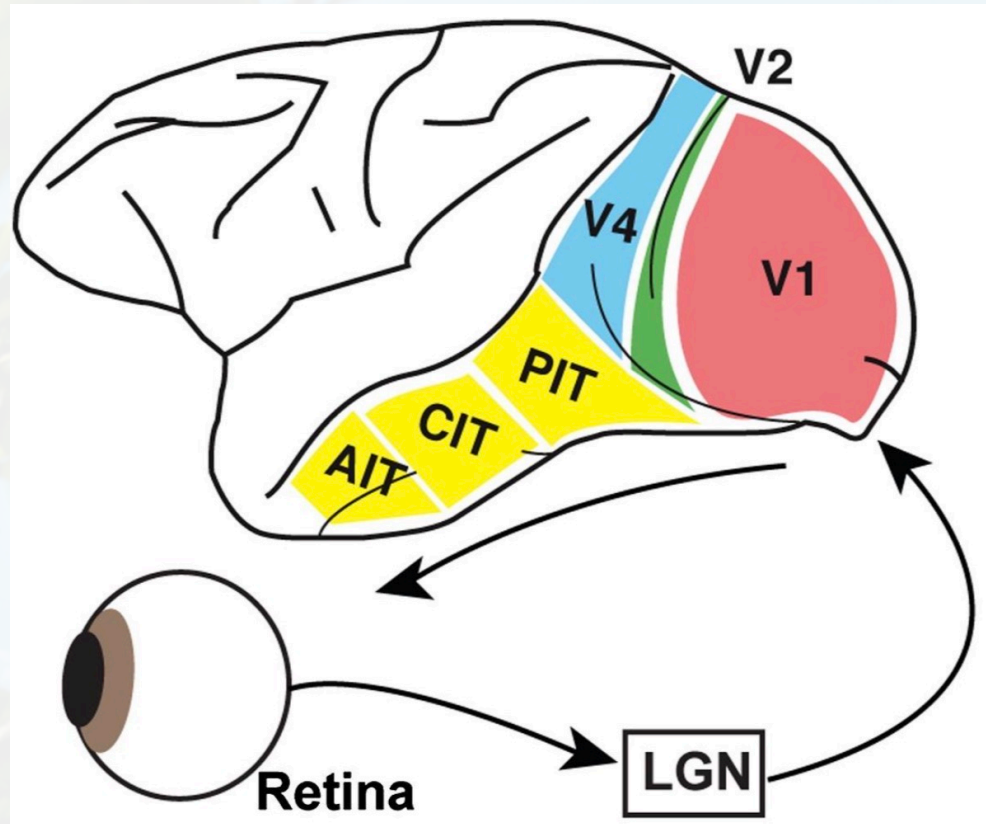


Hubel & Wiesel at Harvard



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V1: low-level processing
(color, orientation, contrast)

V2: intermediate-level
(edge, contour, shape)

V4: high-level (semantic)



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Human vision is superbly efficient

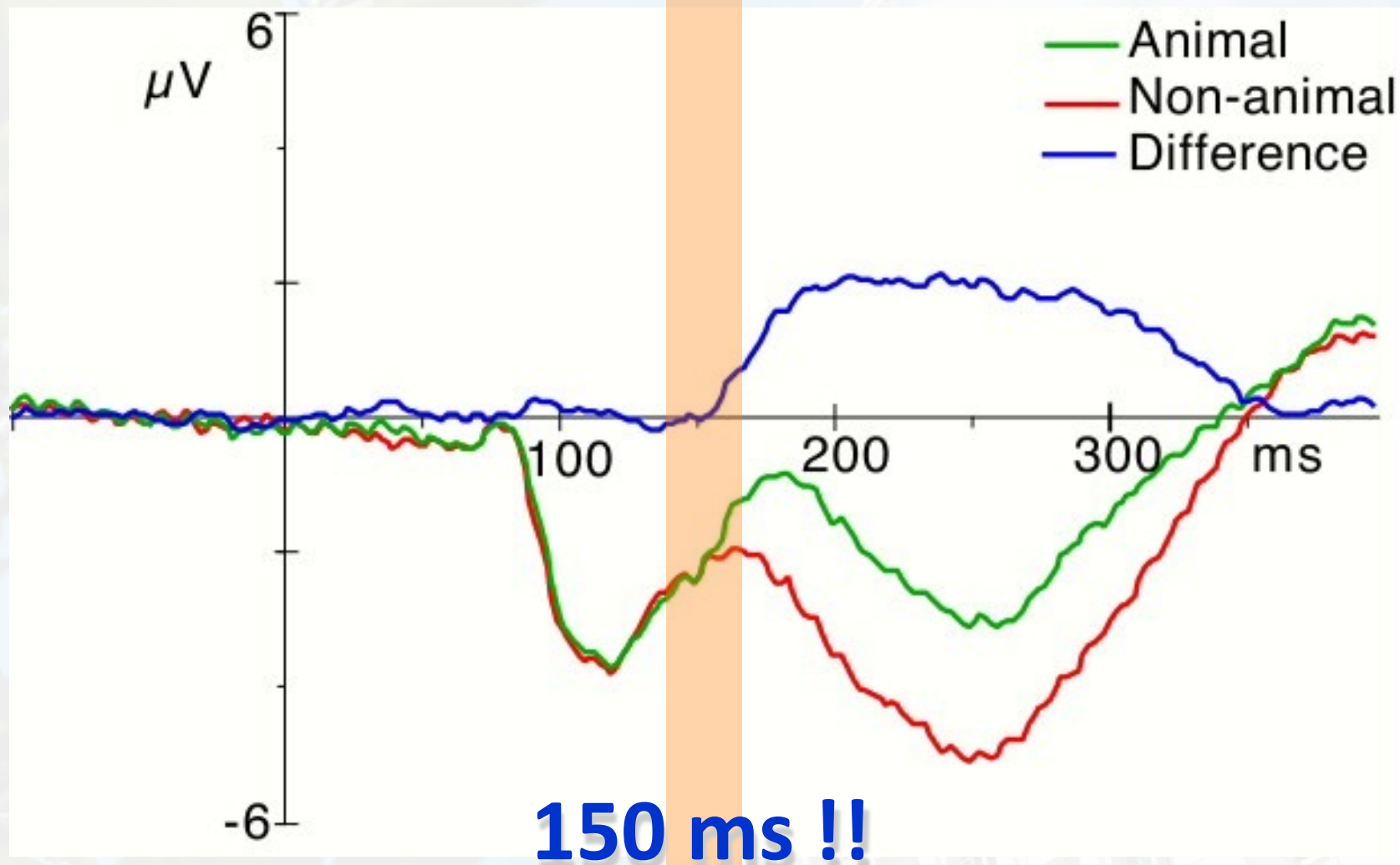


Thorpe, et al. *Nature*, 1996



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Thorpe, et al. *Nature*, 1996



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Fooling humans

However, this speed is obtained at the price of some drawbacks.

- ❑ Changing small irrelevant parts of an image such as water reflection or background can go unnoticed because the human brain focuses on the important parts of an image.
- ❑ If the signal is very close to the background, it can be difficult to detect and segment the relevant part of the image.



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Change blindness



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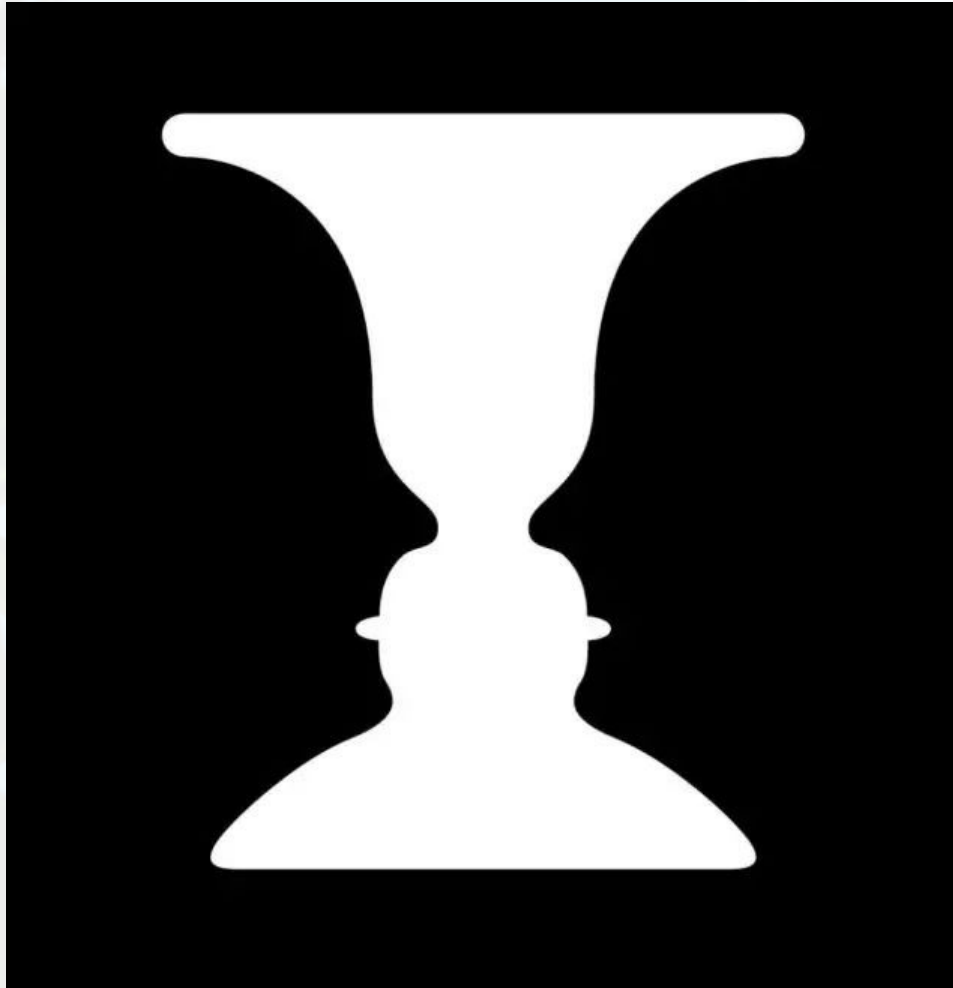
Change blindness



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Segmentation



Rubin's vase
in 1915



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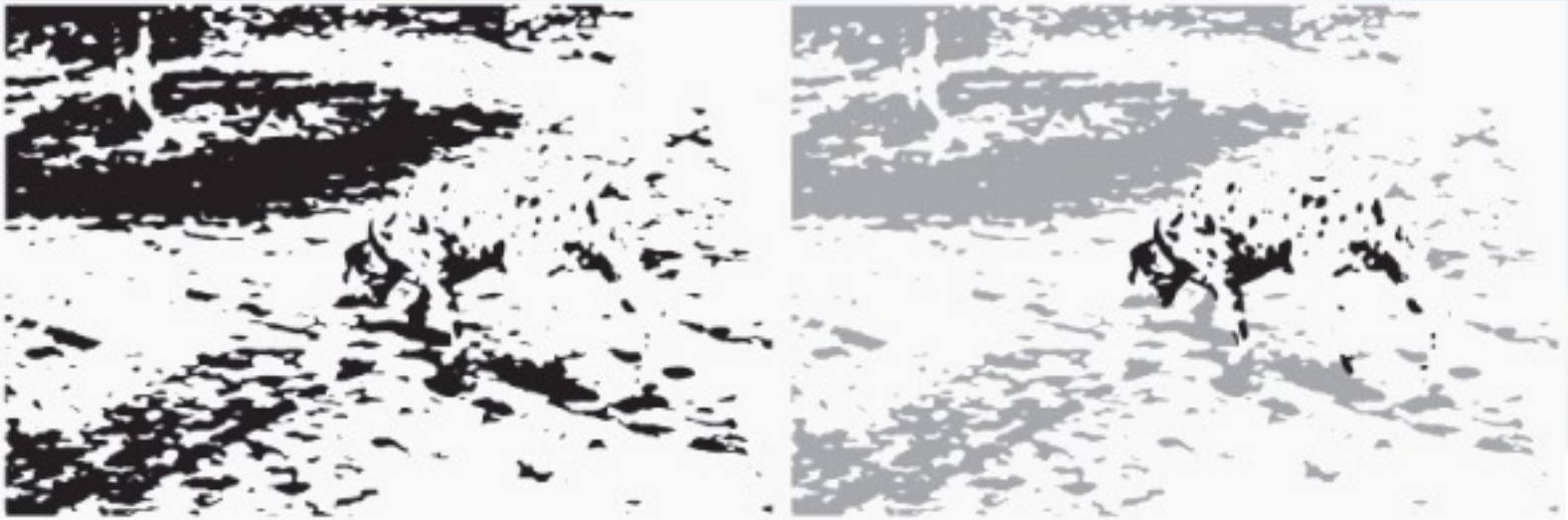
Segmentation

Dalmatian dog



Segmentation

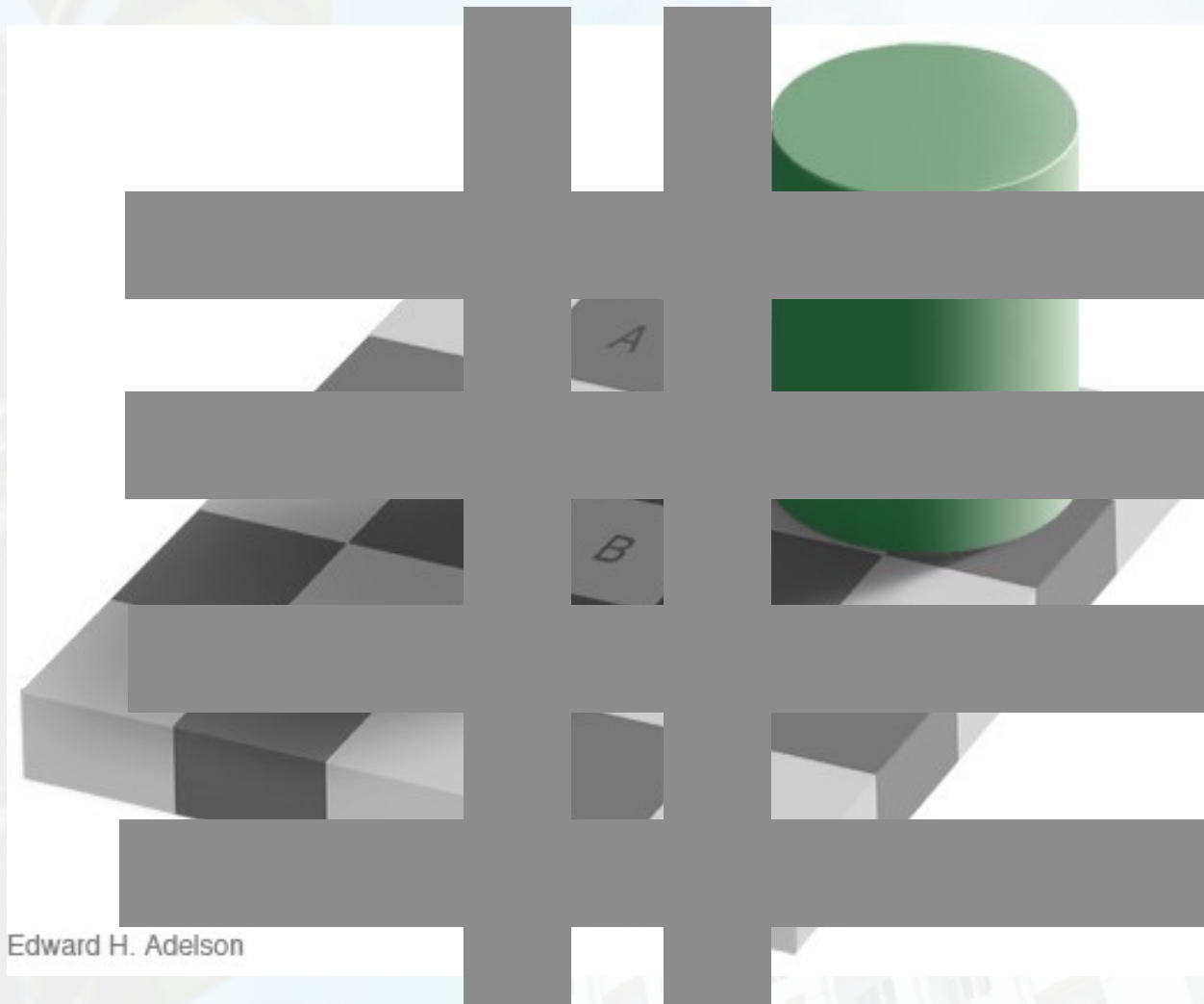
Dalmatian dog



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Perception



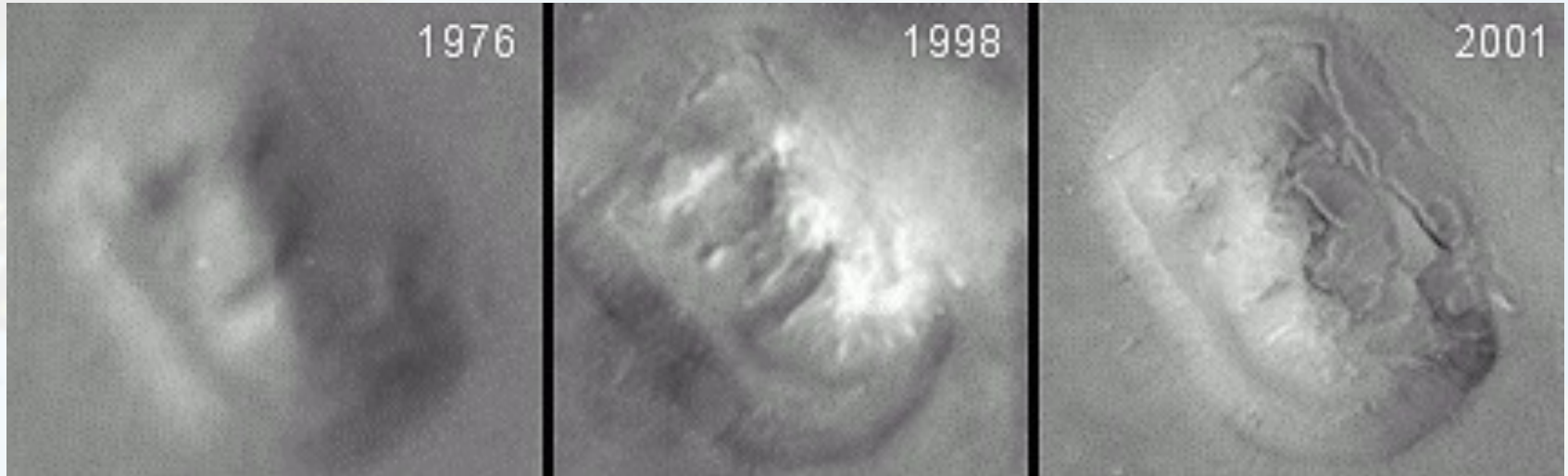
Edward H. Adelson



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Perception



Face on Mars



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Gestalt



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Gestalt



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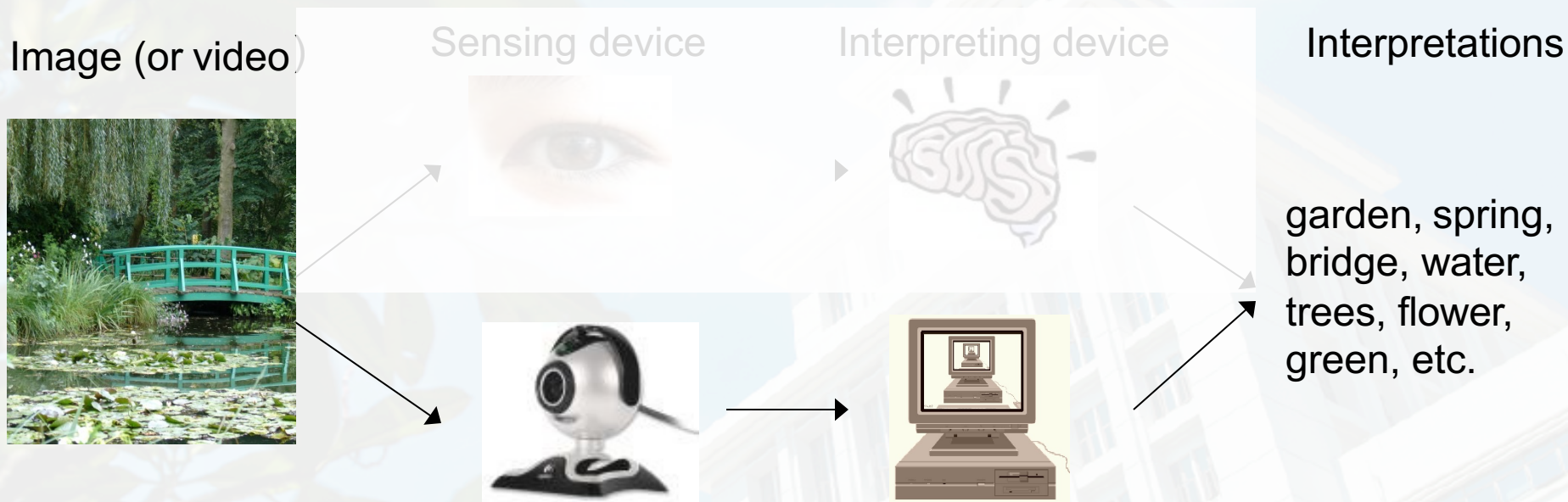
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What is (computer) vision?



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The goal of computer vision

- To bridge the gap between pixels and “meaning”



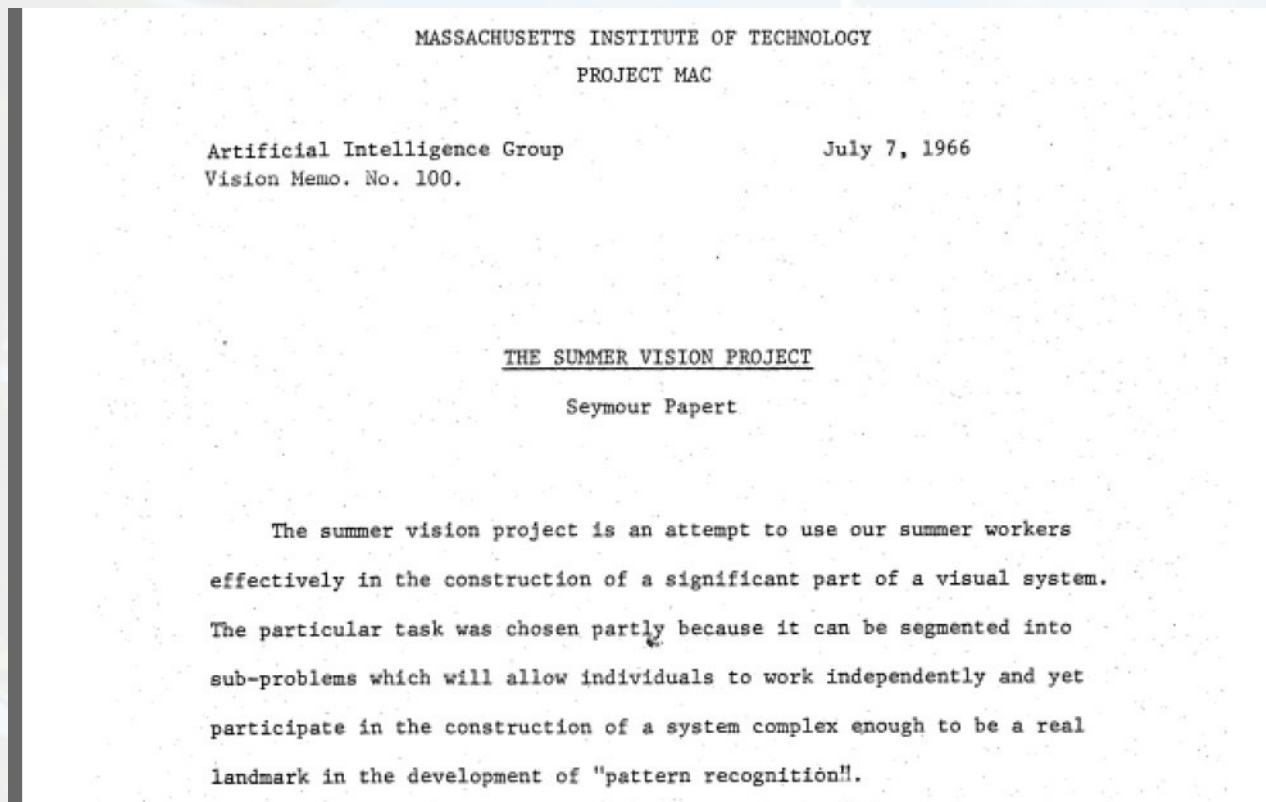
0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0



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Origins of computer vision: an MIT undergraduate summer project



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What kind of information can we extract from an image?

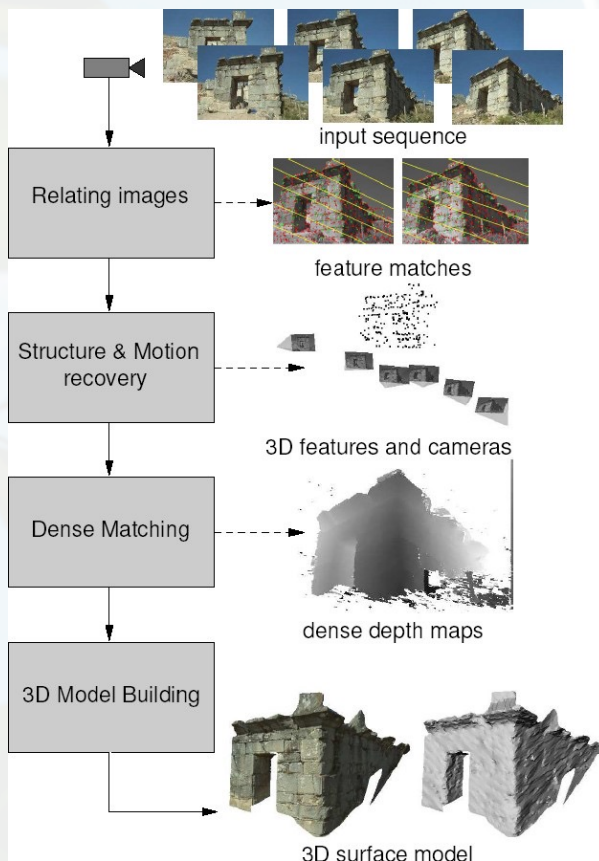
- Measurement
- Semantic information



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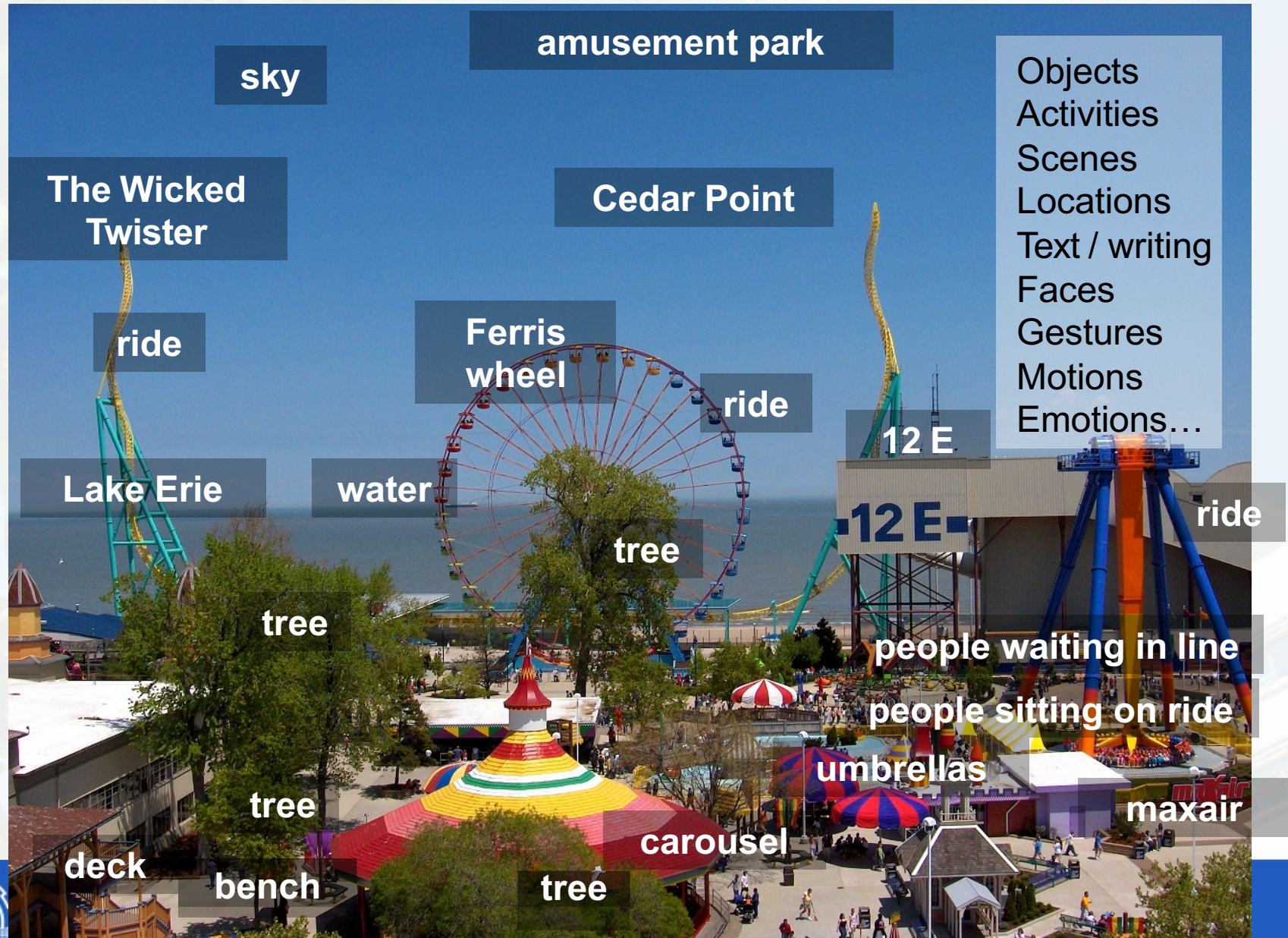
Vision as a measurement device



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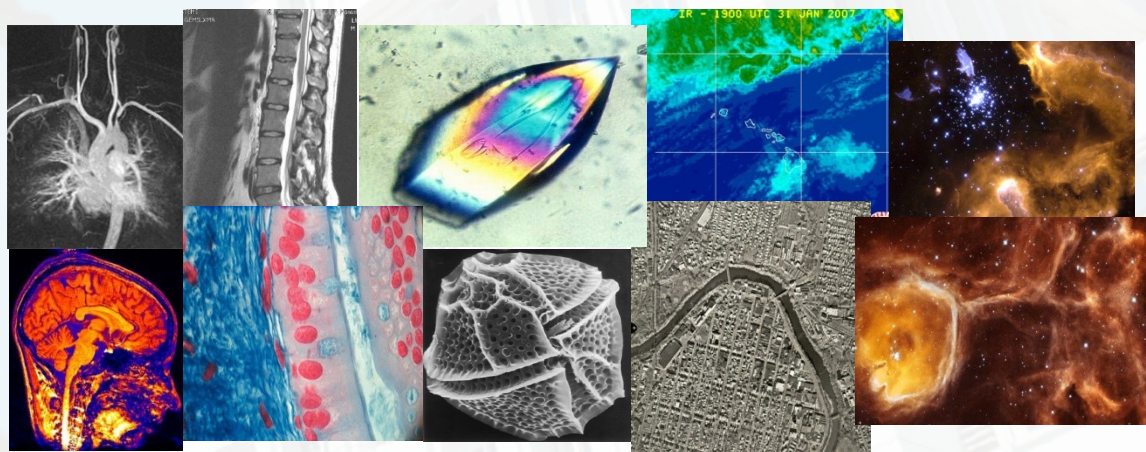
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Vision as a source of semantic information



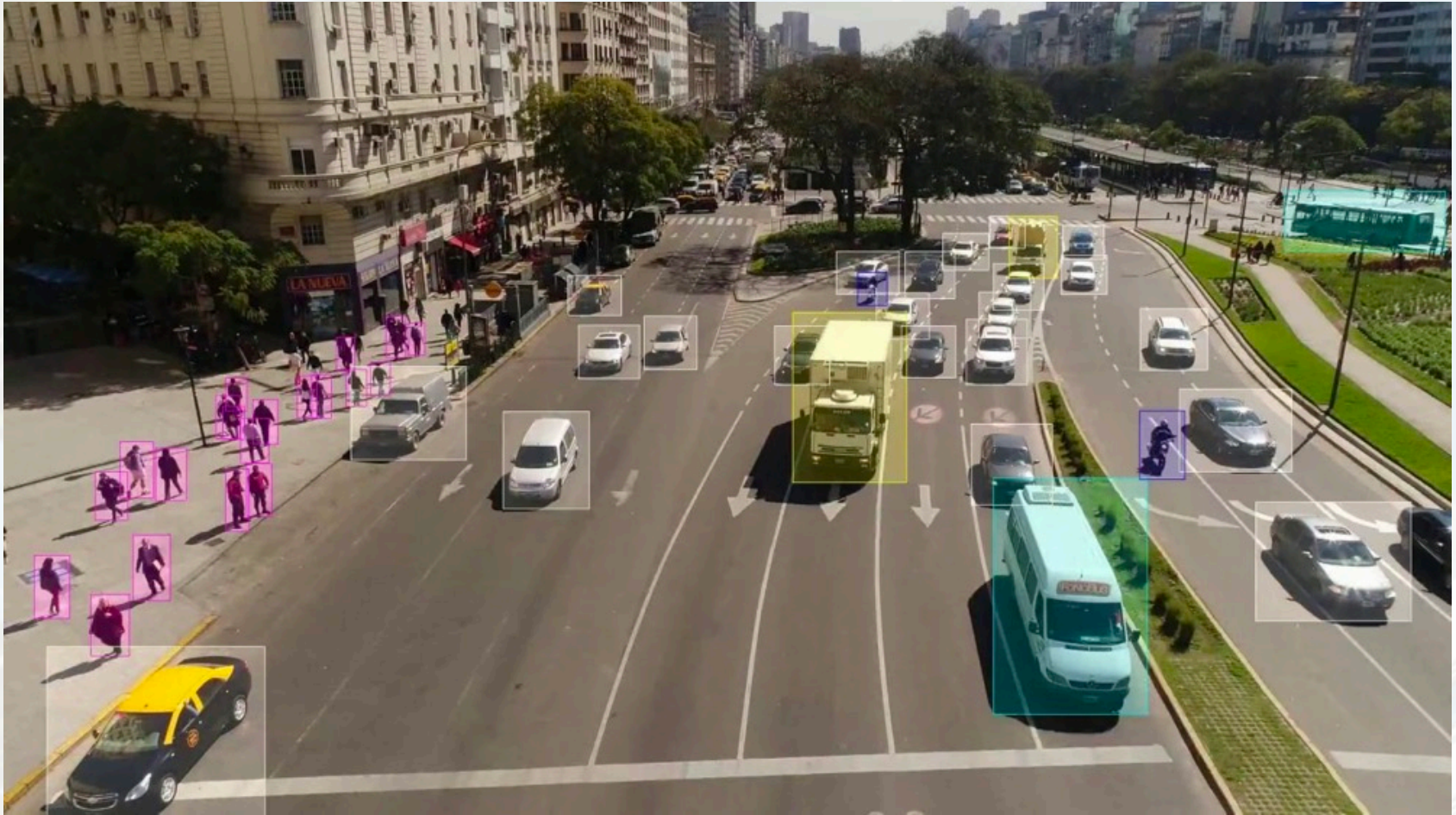
- Vision is useful: Images and video are everywhere!

- Vision is useful: Images and video are everywhere!



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CV in Transportation

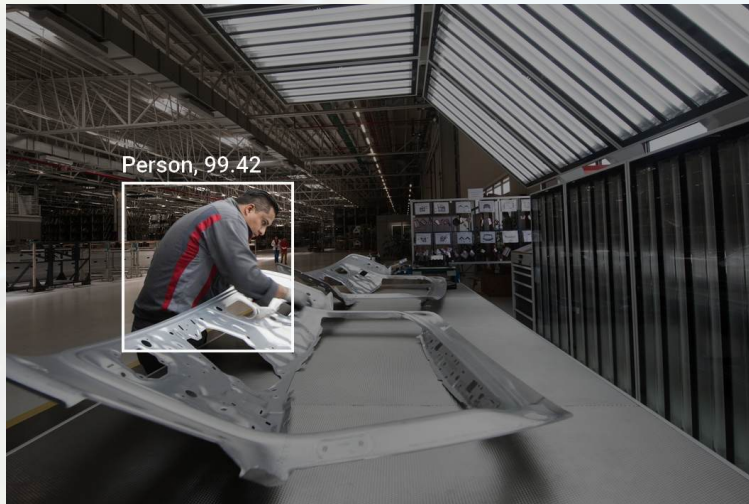


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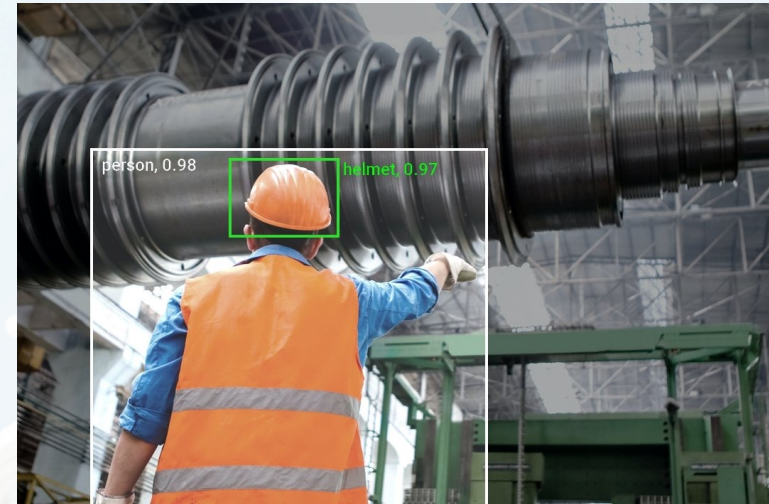
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CV in Manufacturing

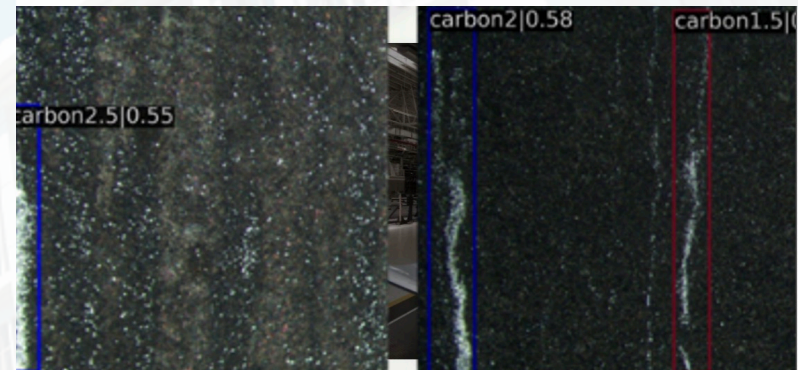
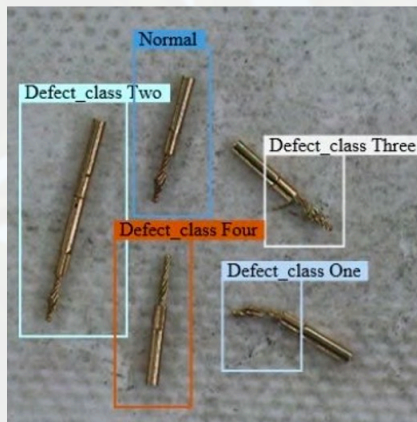
Productivity Analytics



Visual Inspection of Equipment



Quality Management

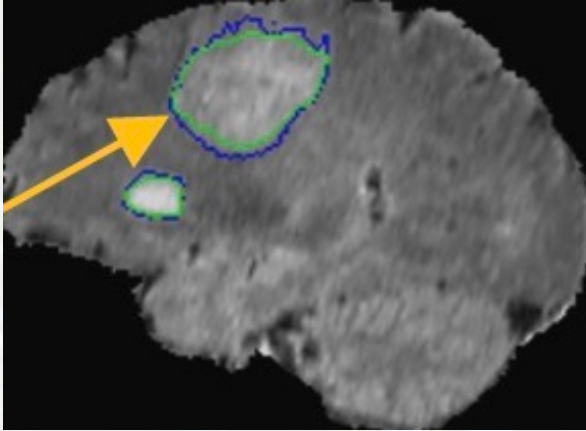


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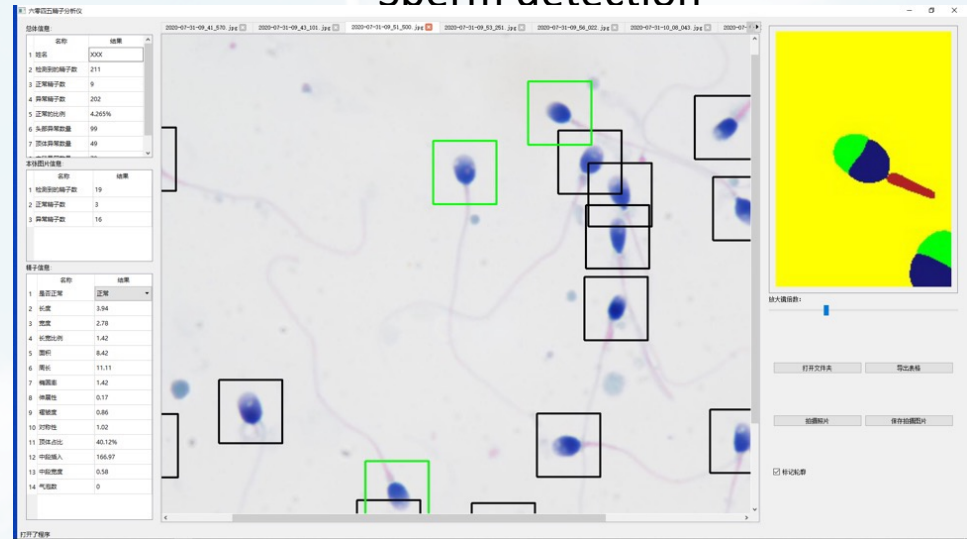
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CV in Healthcare

Tumor detection



Sperm detection



Advanced operating room



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CV in Agriculture

Plant disease detection

Animal monitoring



3 erosomyia_sp

3 erosomyia_sp	93%
2 erosomyia_sp	4%
2 procontarinia_rubus	1%

Automatic harvesting



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CV in Retail

People counting



Product detection



Intelligent assistant



Facial recognition payment

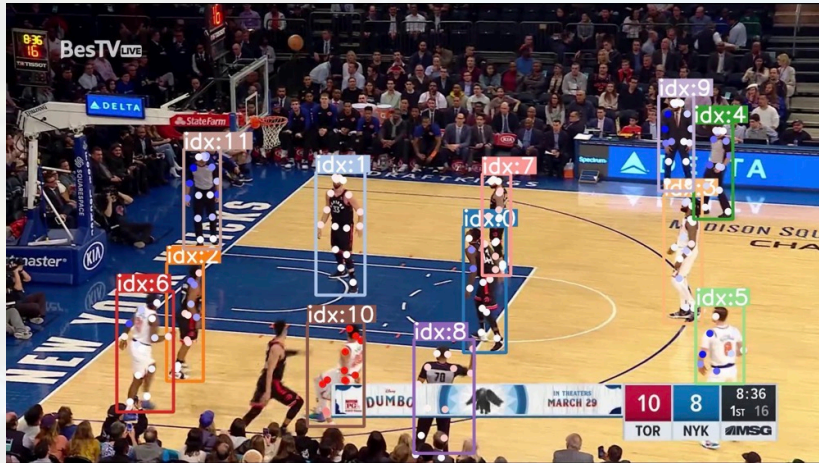


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CV in Sport

Action recognition



Offside detection



Pose estimation



Goal-line technology



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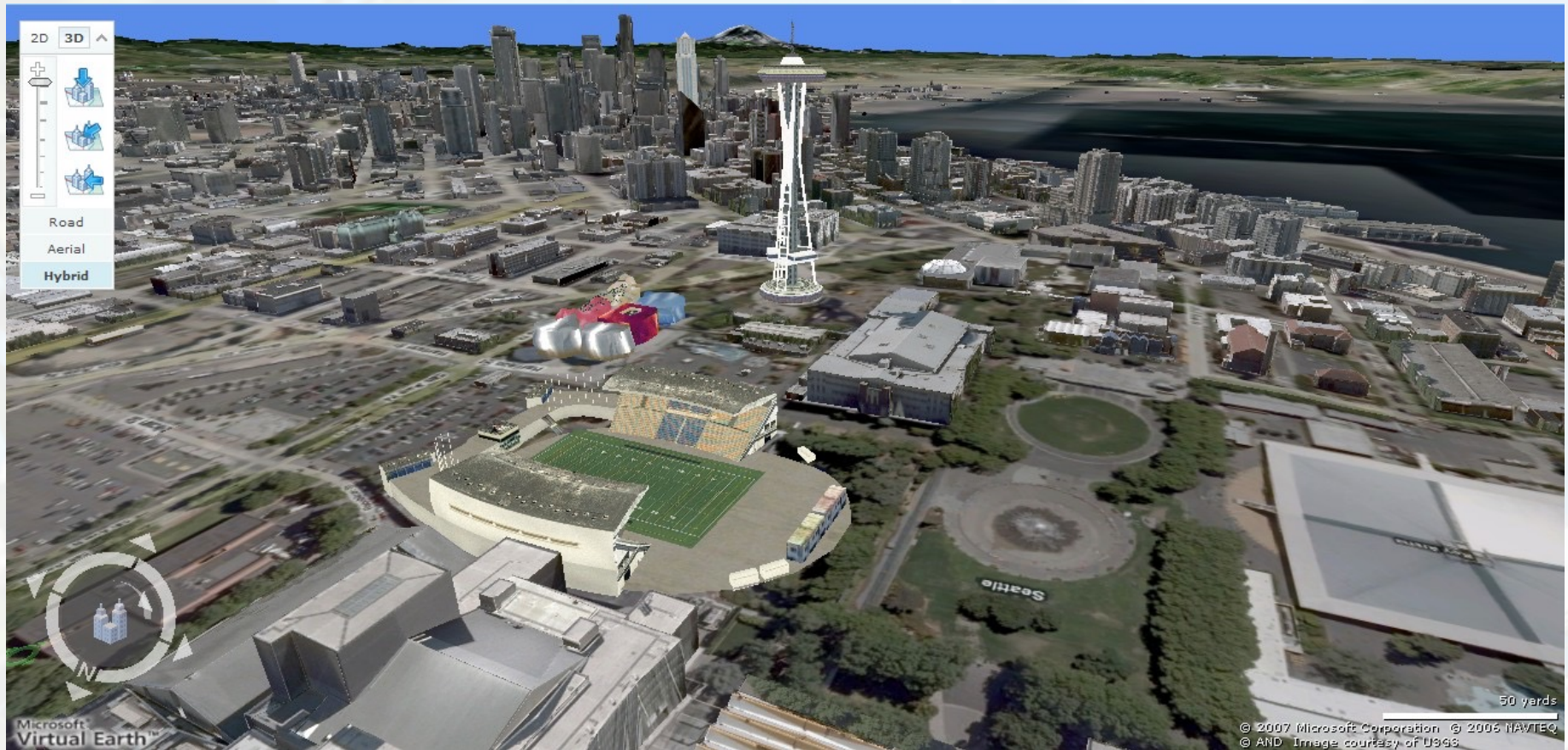
Special effects: shape and motion capture



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3D urban modeling



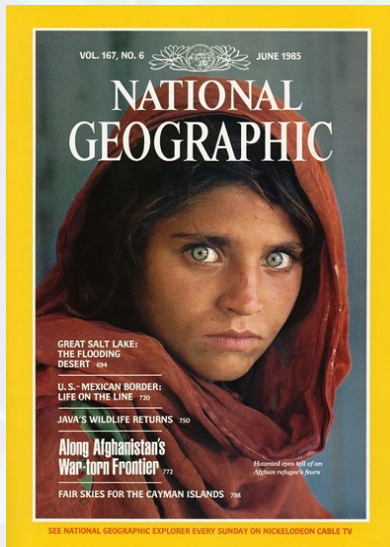
[Bing maps](#), Google Streetview



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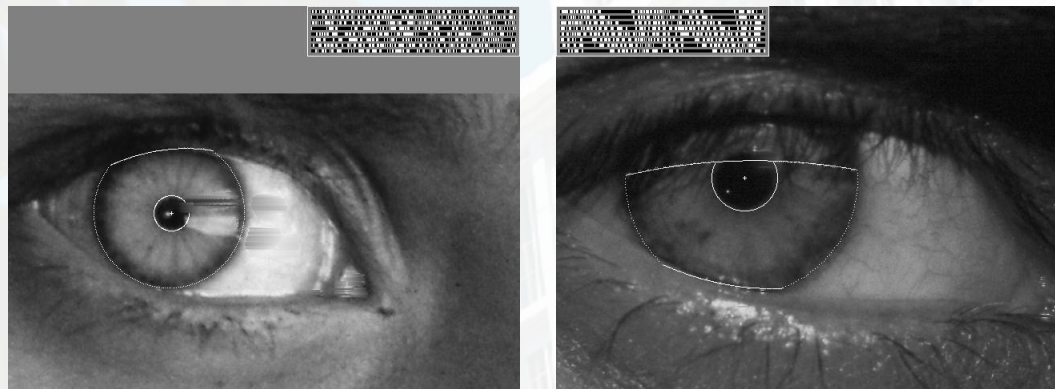
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Biometrics



1984 vs 2002

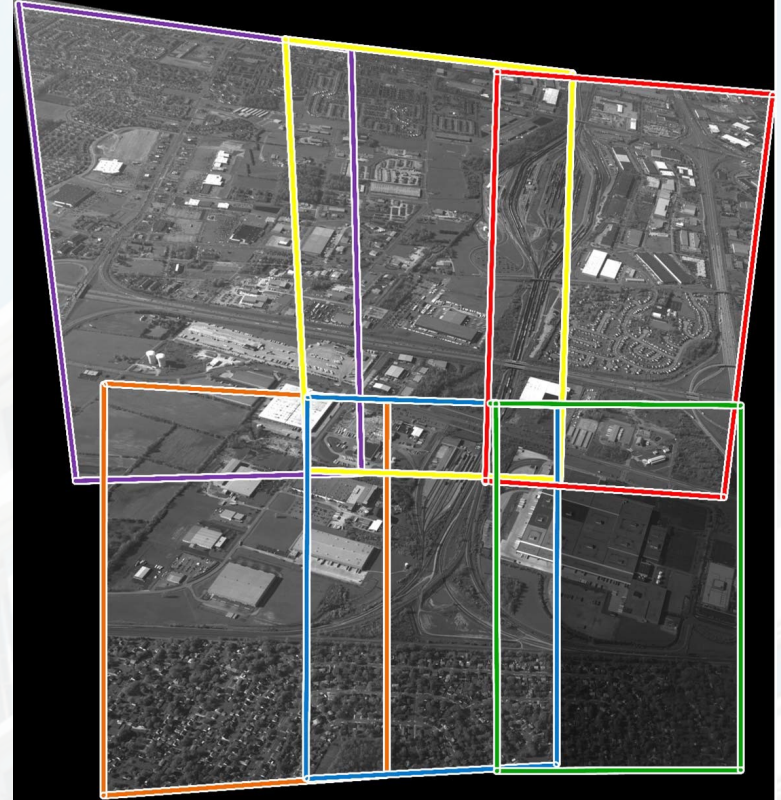
How the Afghan Girl was Identified by Her Iris Patterns



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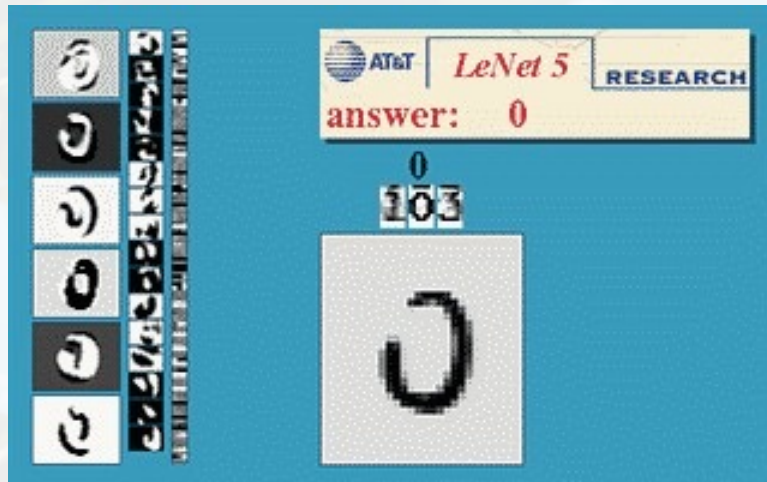
Wide area surveillance



Optical character recognition (OCR)

Technology to convert scanned docs to text

- If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs



License plate readers

http://en.wikipedia.org/wiki/Automatic_number_plate_recognition



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Source: S. Seitz

Today's agenda

- What is computer vision
- **Course overview**



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Course Materials

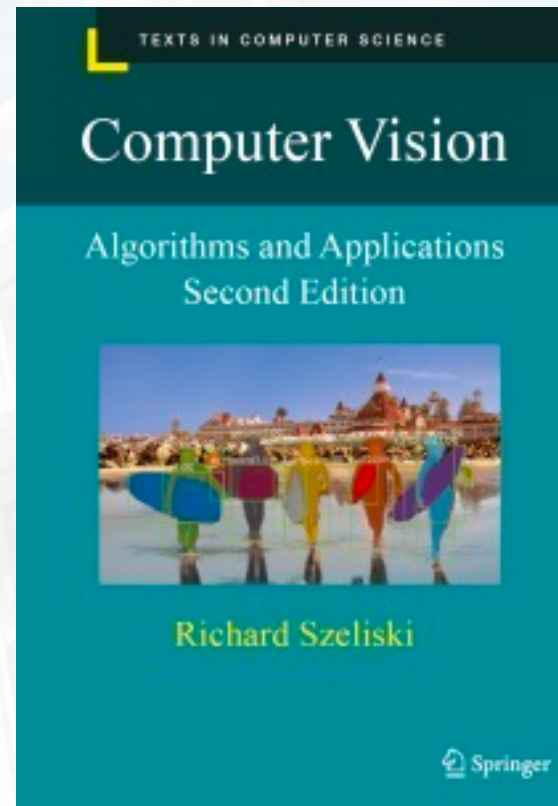
Recommended but not required:

Computer Vision: Algorithms and Applications, 2nd ed. Richard Szeliski

<https://szeliski.org/Book/>

MIT CSAIL: Advances in Computer Vision

<https://advances-in-vision.github.io/>



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Contacting instructor and TA

Instructor: Jingyong Su @ L1501

QQ Group: 982196542

TAs: Haobo Liu, Yi Ma, Zhiqiang Yan



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Topics (16 lectures)

1. Introduction (2)
2. Pixels and Features (2)
3. Regions of Images and Segmentation (2)
4. Object Recognition and Face Detection(2)
5. Neural Networks (2)
6. Recurrent Neural Networks: RNN, LSTM (2)
7. Attention and Transformers (1)
8. Generative Modeling (1)
9. Visual Foundation Model (1)
10. Data-Centric Artificial Intelligence (1)



Your Grades

- Homework Assignment: 60% (12% each)
 - Only the “.ipynb” file
 - Procedures and results are required
- Final Project: 40%



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Image and Transformations



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Types of Images

Binary



Gray Scale



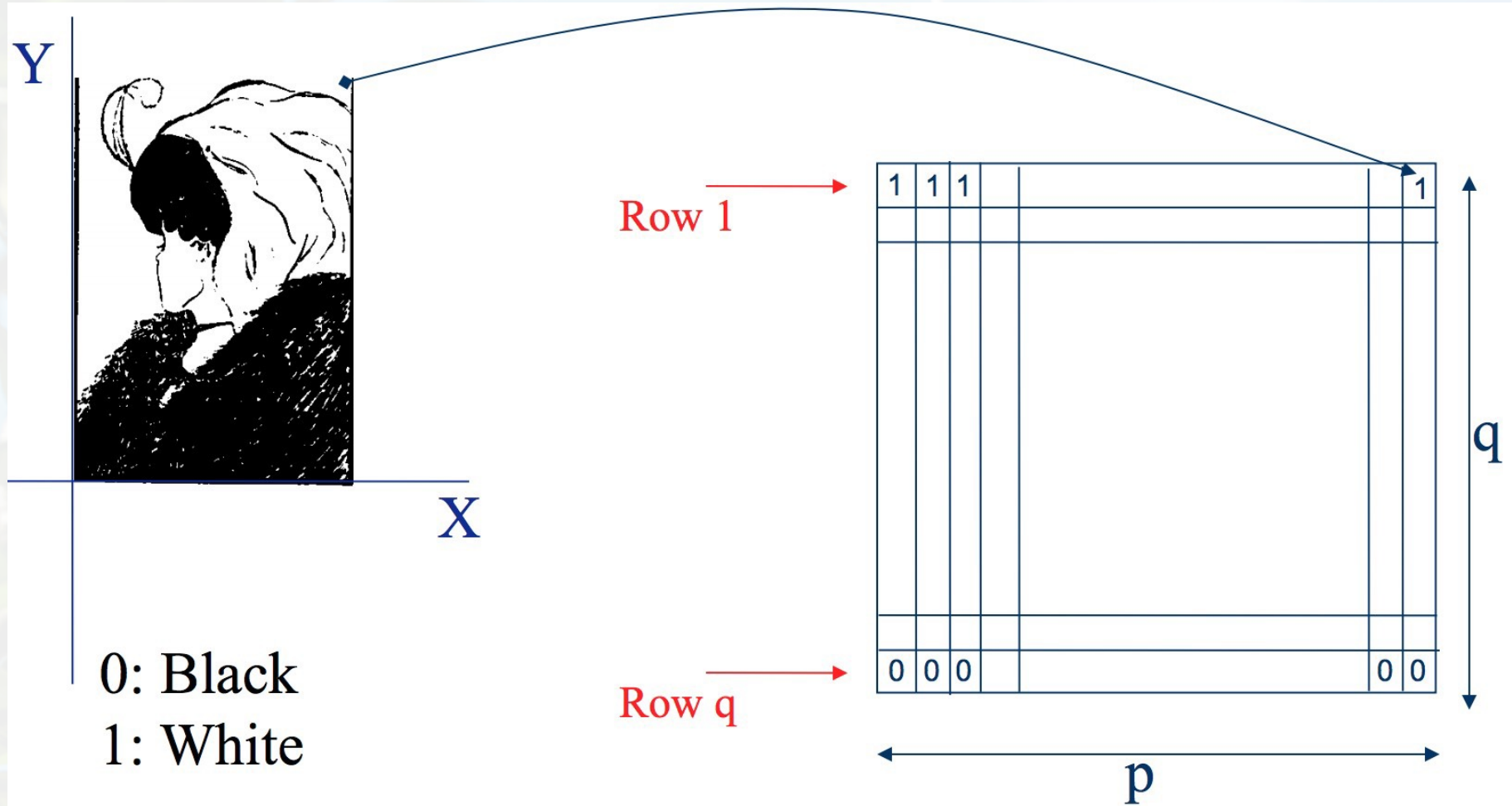
Color



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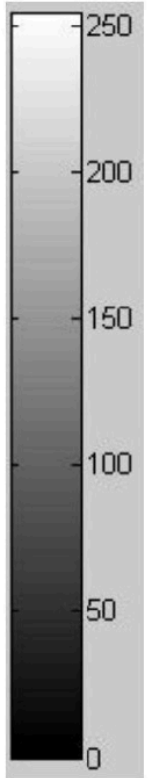
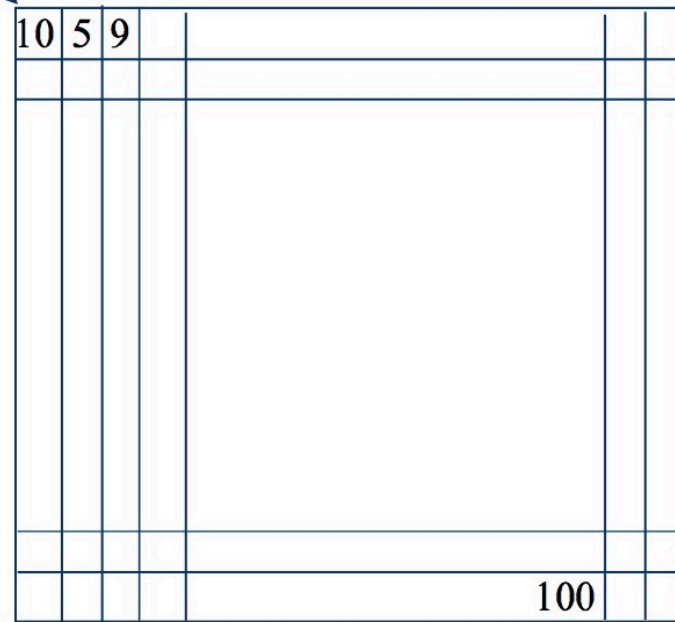
Binary image representation



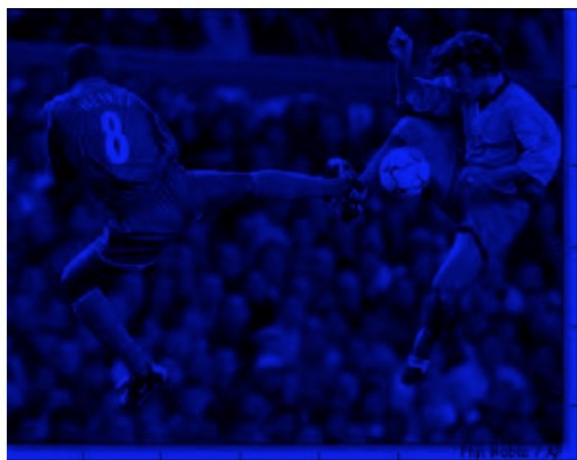
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Grayscale image representation



Color image representation



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Digital Images Are Sampled

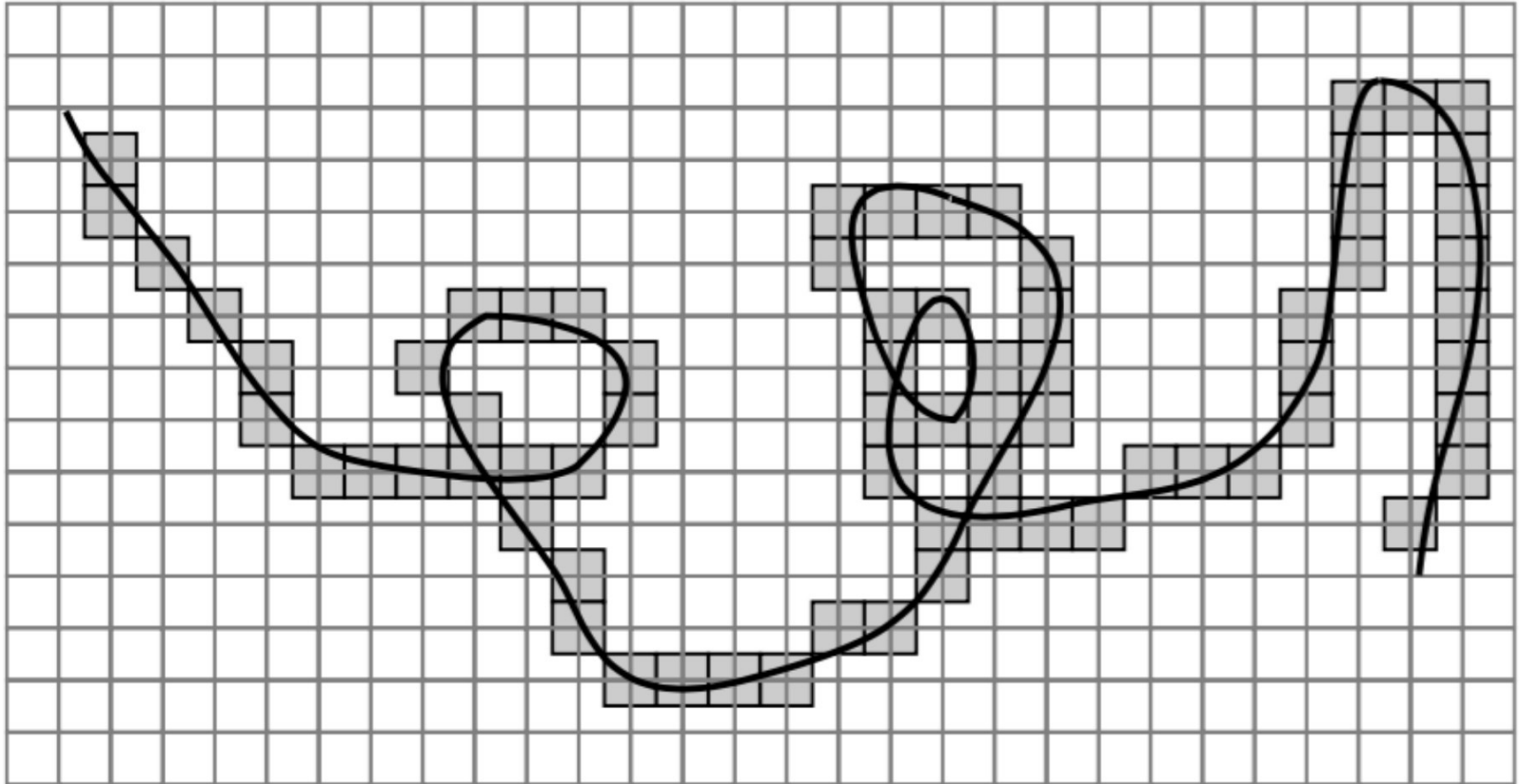
What happens when we zoom into the images we capture?



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Errors due to Sampling



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Resolution

is a **sampling** parameter, defined in dots per inch (DPI) or equivalent measures of spatial pixel density



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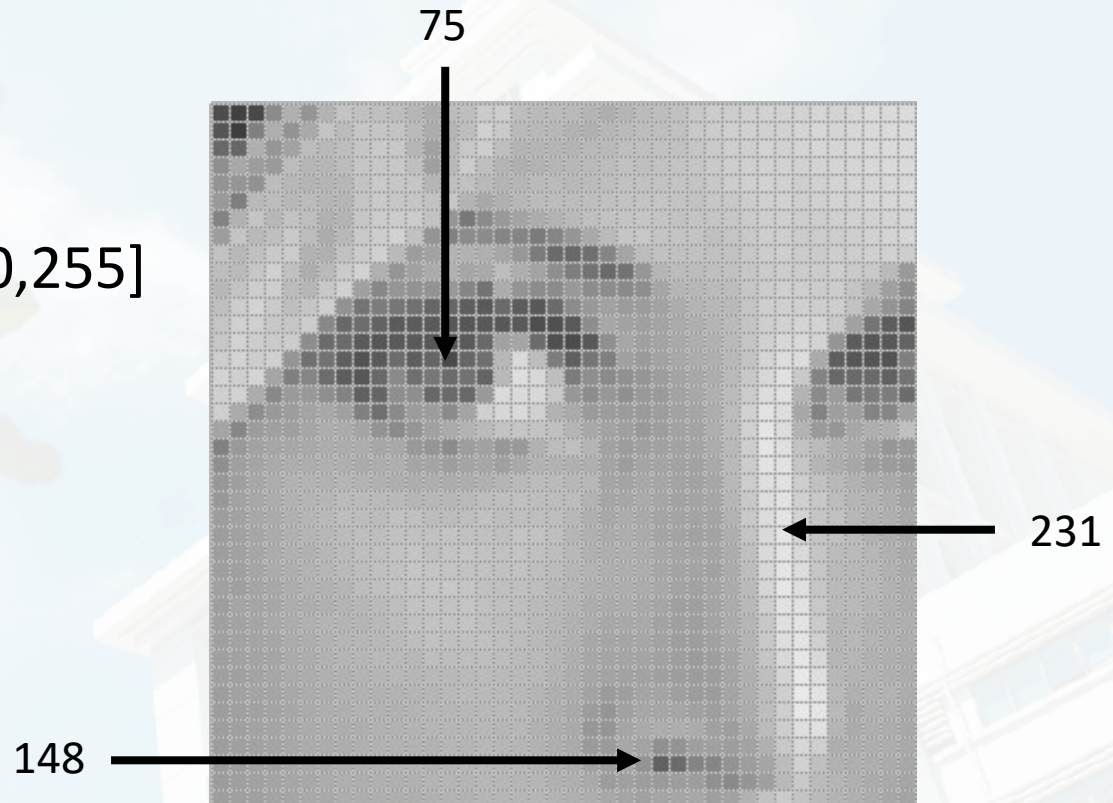
Images are Sampled and Quantized

- An image contains discrete number of pixels

- Pixel value:

- “grayscale”

(or “intensity”): $[0, 255]$



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Images are Sampled and Quantized

- An image contains discrete number of pixels

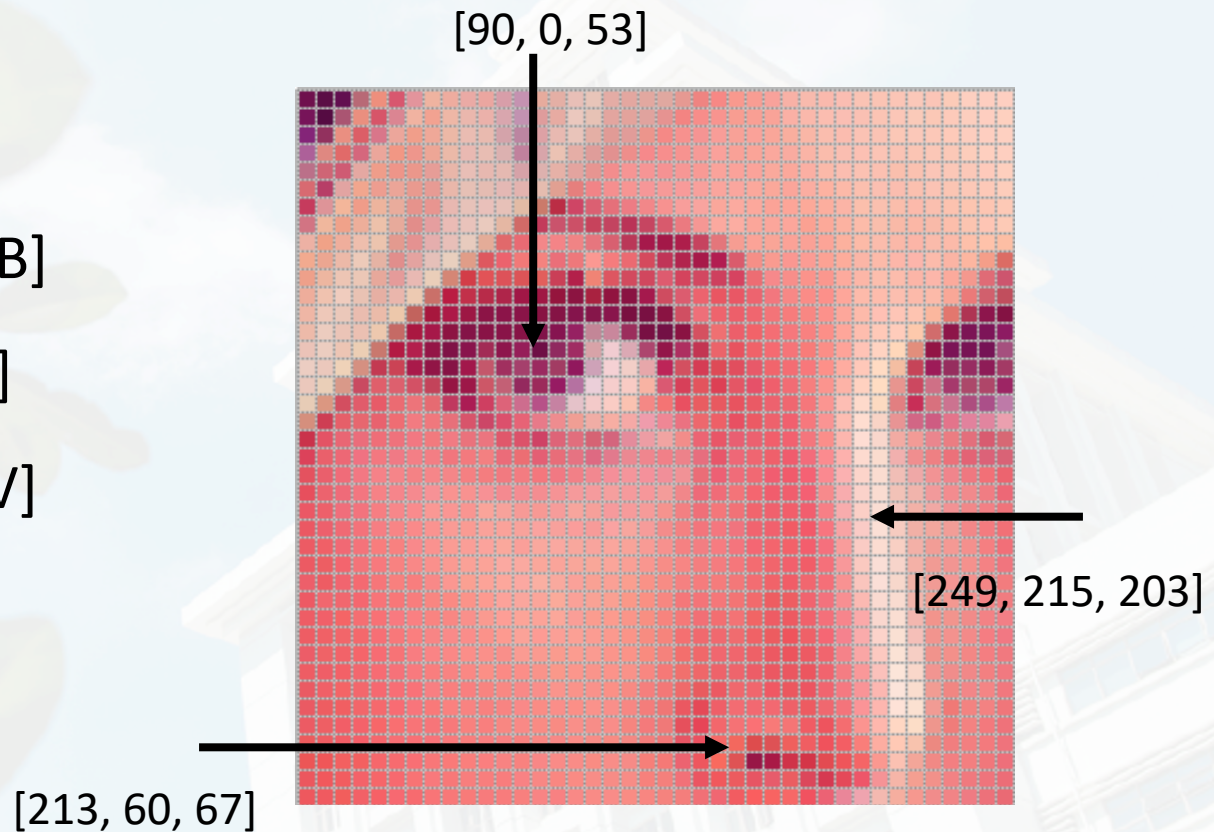
- Pixel value:

- “color”

- RGB: [R, G, B]

- Lab: [L, a, b]

- HSV: [H, S, V]



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Transformations

❖ Translation

❖ Rotation

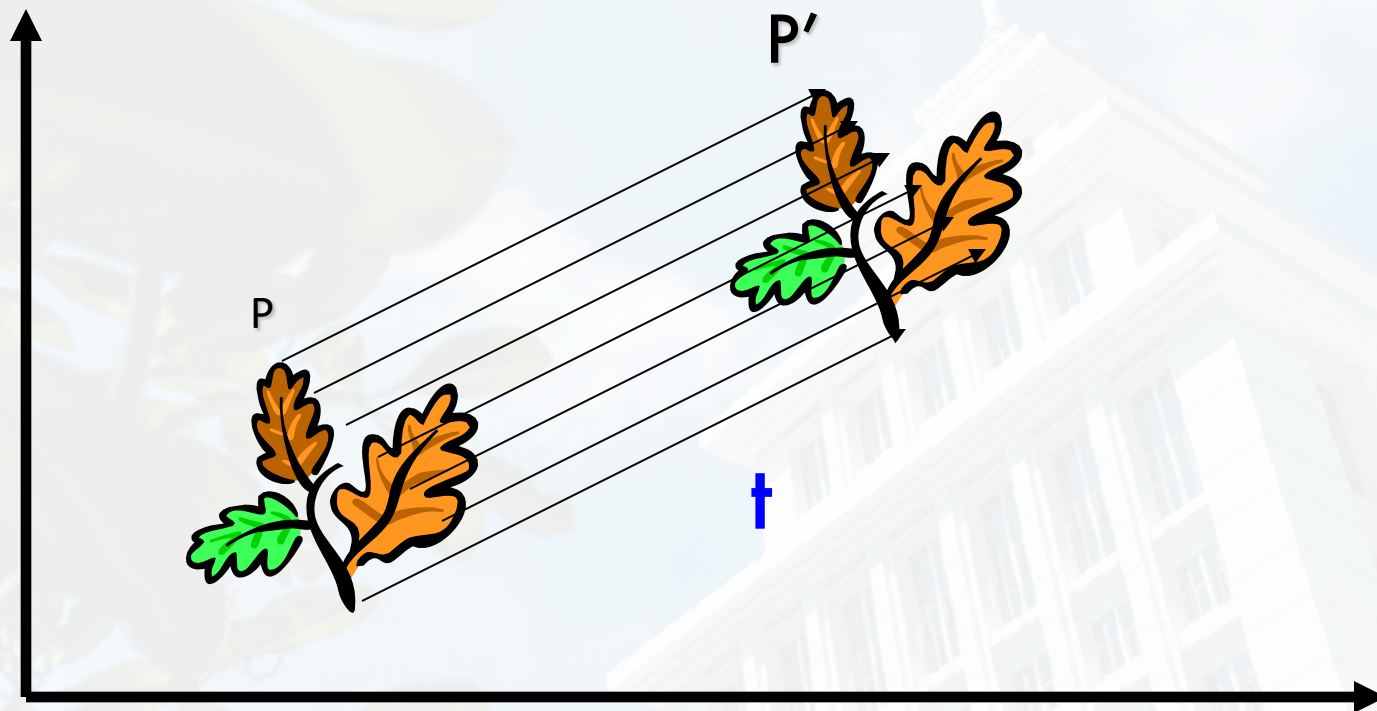
❖ Scaling



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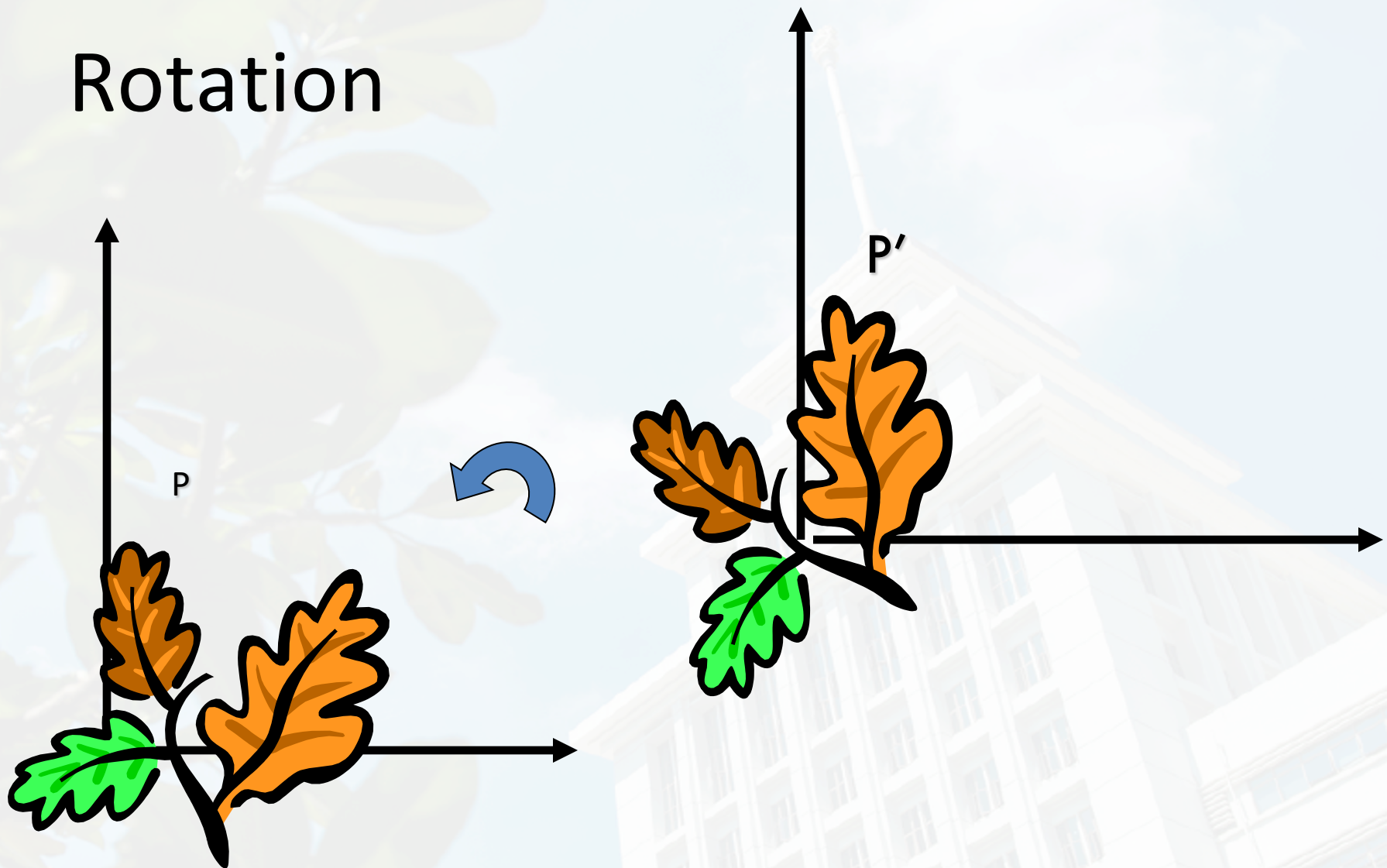
Translation



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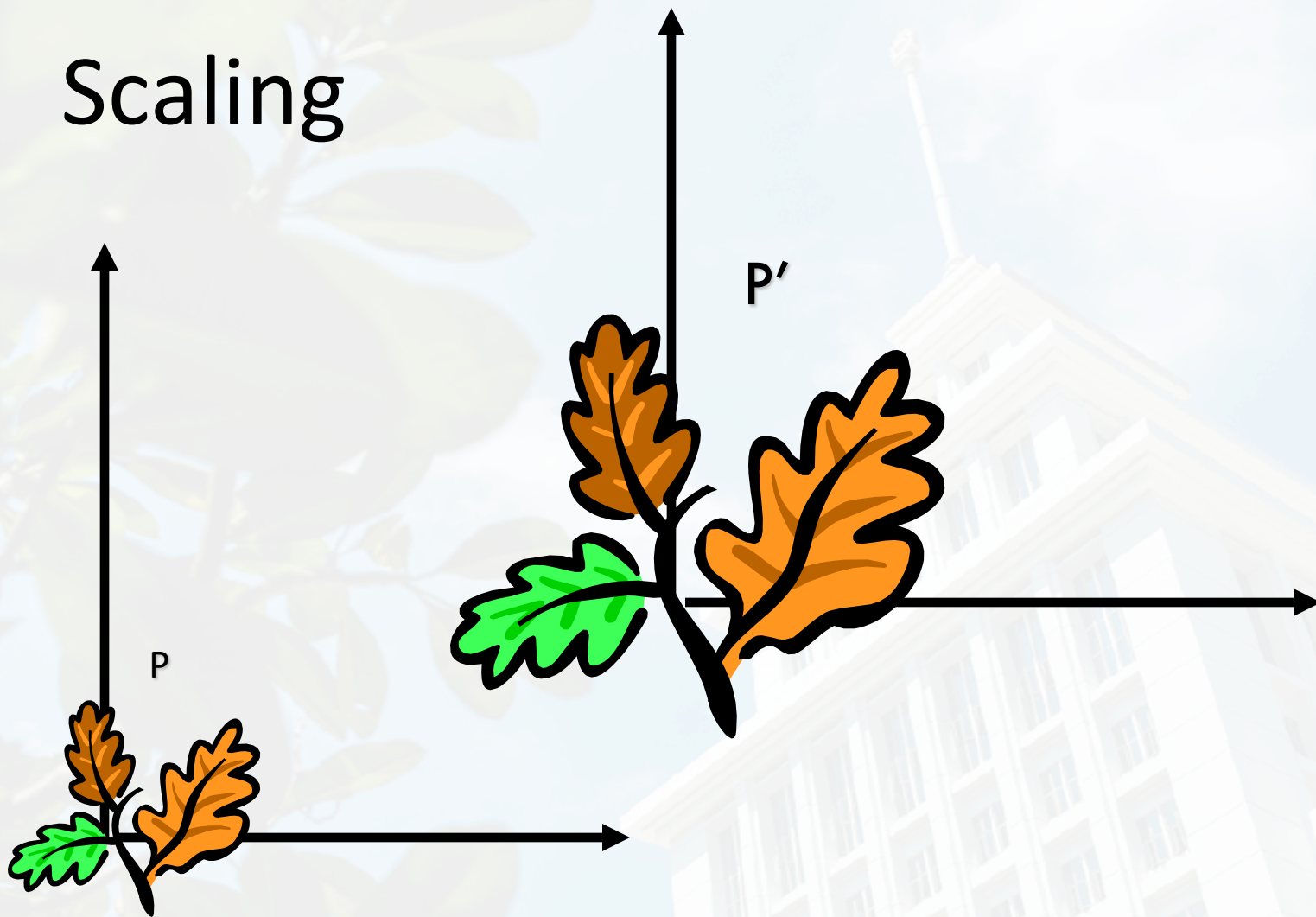
Rotation



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Scaling



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